

Axion Dark Matter Search from Terrestrial Magnetic Fields at Extremely Low Frequencies

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The natural environment of the Earth can act as a sensitive detector for dark matter in ultraviolet axions. When axions with masses in the range of $10^{-15} - 10^{-13}$ eV pass through the Earth, they interact with the global geomagnetic field, generating electromagnetic (EM) waves in the extremely low-frequency range of deci-hertz to several hertz through axion-photon coupling. This paper is one of a series of companion papers for [A. Taruya, A. Nishizawa, and Y. Himemoto, Prog. Theor. Exp. Phys. 2025, 111E01 (2025).] that focus on the data analysis method and the search results for an axion signal. Utilizing the theoretical predictions of axion-induced EM spectra from a companion study, we analyzed the long-term observational data of terrestrial magnetic fields in this frequency band to search for axion-induced signals. Our analysis identified 65 persistent signal candidates with a signal-to-noise ratio greater than 3. Aside from these candidates, we placed a new upper bound on the axion-photon coupling parameter, significantly refining the previous constraint from CAST by at most two orders of magnitude down to $g_{a\gamma} \lesssim 4 \times 10^{-13} \text{ GeV}^{-1}$ for an axion mass of about $3 \times 10^{-14} \text{ eV}$.
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Subject Index B71, C15, E64, E74

1. Introduction

Numerous independent astronomical observations suggest that the majority of matter in the Universe consists of an invisible component known as dark matter (DM). Because DM does not fit within the framework of the standard model of particle physics, its fundamental nature remains one of the most profound enigmas in both cosmology and fundamental physics. Among various candidates for DM, axions were originally proposed to preserve the CP symmetry in quantum chromodynamics (QCD) without requiring fine-tuning of a CP-violating coupling parameter, addressing the so-called strong-CP problem [1–3]. While there exist other axion-like particles [4], which are predicted by string theory and can span a broad mass range, these

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axions and axion-like particles (which we collectively refer to as axions) have long been central to

research as prominent ultralight DM candidates [5–7]. Their possible masses range from 10^{-23} to 1 eV, although recent observations of small-scale structure increasingly disfavor values below a few $\times 10^{-22}$ eV if they make up most of the DM (see, e.g., [8,9] for a review).

The axion DM can couple with electromagnetic (EM) fields, as described by the interaction Lagrangian

$$\mathcal{L}_{\text{int}} = \frac{g_{a\gamma}}{4} a F_{\mu\nu} \tilde{F}^{\mu\nu}, \quad (1)$$

where a is the axion field, $F_{\mu\nu}$ is the EM field strength tensor with its dual given by $\tilde{F}^{\mu\nu} = \epsilon^{\mu\nu\alpha\beta} F_{\alpha\beta}/2$, and $g_{a\gamma}$ represents the coupling coefficient, referred to as the axion–photon coupling. Despite its weak coupling, the interaction offers a direct way to search for axions in laboratory experiments [10,11]. In addition, astronomical observations provide a variety of opportunities to search for the presence of axions through modulation of observed photons, altering the presence of axions changes the lifetimes of astronomical objects, and introducing new phenomena [12,13]. The absence of these signatures, therefore, allows us to constrain the coupling parameter $g_{a\gamma}$.

A novel and independent probe for axion DM is terrestrial EM fields in the extremely low-frequency (ELF) band. Coupled with the static geomagnetic field, coherently oscillating axions permeating the entire Earth can generate monochromatic EM waves at a frequency corresponding to the axion mass m_a . The use of such an environment was investigated in Ref. [14], which obtained a constraint on $g_{a\gamma}$ in the mass range of $2 \times 10^{-18} \text{ eV} \lesssim m_a \lesssim 7 \times 10^{-17} \text{ eV}$ from data from the magnetometer network on Earth. More recently, the heavier mass range, $m_a \lesssim 4 \times 10^{-15} \text{ eV}$, has been constrained with the high-sampling measurement data of terrestrial magnetic fields [15]. There are other efforts constraining the coupling with direct measurements in the mass range 10^{-15} – 10^{-12} eV [16–20].

In this paper, we focus on EM fields at the frequencies of deci-Hz to several Hz, corresponding to an axion mass of 10^{-15} – 10^{-13} eV. For the frequency band, the wavelength can be comparable to the Earth circumference. In contrast to the frequency range of $f \lesssim 1$ Hz studied in Refs. [14,15], EM waves generated near the Earth’s surface at certain frequencies resonate between the Earth’s surface and the ionosphere and show spectral peaks in the frequency band of $1 \text{ Hz} \lesssim f \lesssim 100 \text{ Hz}$. Therefore, a proper theoretical modeling of the atmospheric cavity resonator that accounts for altitude-dependent conductivity is required to predict a finite axion-induced EM amplitude.

This paper is one of a series of papers focusing on the data analysis method and search results for an axion signal, also placing a constraint on the coupling parameter $g_{a\gamma}$ using long-term monitoring data of terrestrial magnetic fields. Alongside this work, a separate paper presents the theoretical formulation and prediction of induced EM waves [21], while a letter highlights the key findings from both theoretical and data analysis studies [22]. We show that the most stringent bound on the axion–photon coupling among direct observational means is obtained across the mass range of 1×10^{-15} – 1×10^{-13} eV.

The organization of the paper is as follows. In Sec. 2, we summarize the theoretical prediction of the axion-induced EM spectrum, which was derived in the companion paper [21]. Our data analysis method is described in Sec. 3, and the data used and the results obtained (an axion signal estimator, SNR, and axion signal candidates) are presented in Sec. 4. From the amplitude

distribution of an axion signal estimator, we obtain an observational constraint on the axion–photon coupling strength in Sec. 5. Finally, Sec. 6 is devoted to the conclusions. In this paper, we adopt the units of $c = \hbar = 1$.

2. Axion-induced signal in terrestrial B -fields

In the presence of axions, the Maxwell equations are modified, and the effective charge and current appear through the axion–EM coupling [10,21]:

$$\nabla \cdot (n^2 \mathbf{E}) = g_{a\gamma} \nabla \cdot (a \mathbf{B}), \quad (2)$$

$$\nabla \cdot \mathbf{B} = 0, \quad (3)$$

$$\nabla \times \mathbf{E} + \partial_t \mathbf{B} = 0, \quad (4)$$

$$\nabla \times \mathbf{B} - \partial_t (n^2 \mathbf{E}) = -g_{a\gamma} [\nabla \times (a \mathbf{E}) + \partial_t (a \mathbf{B})], \quad (5)$$

where n is the complex refractive index.

If the axions constitute DM, they possess a non-relativistic velocity of approximately $v_{\text{DM}} \sim 10^{-3}$. In this case, terms involving the spatial gradient, $|\nabla a| \sim m_a v_{\text{DM}} a$, are generally negligible compared with those involving the time derivative, $|\partial_t a| \sim m_a a$. As a result, the primary modification in Eq. (5) arises solely through the effective current, which is expressed as $\mathbf{J}_{\text{eff}} = -g_{a\gamma} (\partial_t a) \mathbf{B}$.

The axion field equation,

$$\partial_t^2 a - \nabla^2 a + m_a^2 a = -g_{a\gamma} \mathbf{E} \cdot \mathbf{B}, \quad (6)$$

is in general solved with the Maxwell equations above. However, because the backreaction to axion amplitude is sufficiently small in the current situation, the source term in Eq. (6) is negligible. Using the same approximation that the spatial gradient is much smaller than the time derivative, we have a simple solution that oscillates coherently with m_a :

$$a = a_0 e^{-im_a t}. \quad (7)$$

Therefore, the axions are predominantly coherent and produce an alternating current with a frequency of

$$f_a = \frac{m_a}{2\pi} \simeq 2.4 \left(\frac{m_a}{10^{-14} \text{ eV}} \right) \text{ Hz} \quad (8)$$

when coupled with a static magnetic field. The amplitude a_0 is related to the local DM density ρ_{DM} by

$$a_0 = \frac{\sqrt{2\rho_{\text{DM}}}}{m_a}. \quad (9)$$

Because the de Broglie wavelength of the axion is estimated to be

$$\lambda_{\text{DM}} = \frac{1}{m_a v_{\text{DM}}} \sim 1 \text{ AU} \left(\frac{10^{-14} \text{ eV}}{m_a} \right) \left(\frac{10^{-3}}{v_{\text{DM}}} \right), \quad (10)$$

we can take the field value a_0 to be constant over the Earth's geometry for the axion mass of interest.

In the presence of an alternating current, the modified Ampère–Maxwell law suggests that monochromatic EM waves are produced with a frequency determined by Eq. (8). The bandwidth of the produced EM waves is estimated to be $\Delta f / f_a \sim v_{\text{DM}}^2 \sim 10^{-6}$. Thus, the EM waves induced by the coherent axion manifest as a prominent and sharp spike in frequency, and the signal will exist steadily over the coherence time given by $\lambda_{\text{DM}} / v_{\text{DM}}$.

As a representative static magnetic field, we consider the geomagnetic field, $\mathbf{B}_{\text{geo}}$, characterized predominantly by a dipole configuration, with a typical strength of $|\mathbf{B}_{\text{geo}}| \sim 25 - 65 \mu\text{T}$ [23], and give a quantitative prediction of the axion-induced EM waves. Although our primary interest lies in the magnetic fields rather than in the electric fields, the electric field equation is easier to handle when imposing boundary conditions. Moreover, once we obtain a solution for the electric fields, a solution for the magnetic fields can easily be derived from Eq. (4).

The electric field equation is obtained by substituting Eq. (5) into the rotation of Eq. (4). Replacing the magnetic field $\mathbf{B}$ in the electric field equation with the geomagnetic field $\mathbf{B}_{\text{geo}}$, we obtain

$$\nabla^2 \mathbf{E} - n^2 \partial_t^2 \mathbf{E} - \nabla(\nabla \cdot \mathbf{E}) = -g_{a\gamma} (\partial_t^2 a) \mathbf{B}_{\text{geo}}. \quad (11)$$

For the geomagnetic field $\mathbf{B}_{\text{geo}}$, we adopt the IGRF-13 model of the Earth's magnetic field [23]¹, which provides the harmonic coefficients up to $\ell = 13$. Note that by allowing the refractive index to vary spatially, the electric field is not divergence-free even in the absence of charge [see Eq. (2)], and thus the third term on the left-hand side in Eq. (11) becomes non-vanishing. In this paper, we are particularly interested in the axion-induced EM waves in the presence of the geomagnetic field. We shall thus consider a monochromatic EM wave with the frequency $f_a = m_a/(2\pi)$ in Eq. (8). In this case, the refractive index is related to the conductivity σ through

$$n^2(r) = 1 + i \frac{\sigma(r)}{2\pi f_a}. \quad (12)$$

Here we consider the conductivity as a function of the radius r from the center of the Earth. It is known that the radial profile of atmospheric conductivity significantly affects the amplitude and spectral features of the ELF EM waves. The conductivity also varies in time owing to diurnal and seasonal effects and space-weather conditions. Such temporal variations in the conductivity can in principle affect our results. However, based on the study of the day–night variation of conductivity profiles in the companion paper [21] for the theoretical modeling of axion-induced magnetic fields, we found that the magnetic-field signal varies by less than 10% for masses below $3 \times 10^{-13} \text{ eV}$. For long-term temporal variations, at altitudes of the order of 100 km, the conductivity can undergo several-fold variations over the decadal timescales associated with solar activity [24]. Although the conductivity that contributes to our results is at altitudes between 50 and 110 km and would not be affected significantly by space-weather conditions, we can expect, based on the study of the day–night variation above, that the effect of the long-term temporal variations on our results would be less than 10%. Therefore we conclude that the effects of temporal variations on our results are insignificant and we assume a static conductivity profile in Eq. (12).

Because the Earth's interior is highly conductive, we can treat the Earth's surface as a perfect conducting boundary and impose the condition that the non-radial direction of the electric fields should vanish, i.e., $\mathbf{E}_{\parallel} = 0$ at the Earth's radius, $r = R_E$. On the other hand, regarding the atmospheric conductivity, while it is small ($\sim 10^{-12} \text{ S/m}$) near the Earth's surface, it increases significantly to $\sim 10^{-2} \text{ S/m}$ at the altitude of the lower ionosphere, $r \sim R_E + 100 \text{ km}$; see e.g. [25–27]. Although this is still lower than the typical conductivity inside the Earth ($\sim 10 \text{ S/m}$), the EM fields exhibit a diffusion-like behavior and they are expected to decay rapidly above the

¹In the later analysis, we do not take into account the geomagnetic secular variation, which is the variation of the field strength on timescales longer than a year, because the variation is typically less than 1% and is negligible for the analysis and results in this paper.

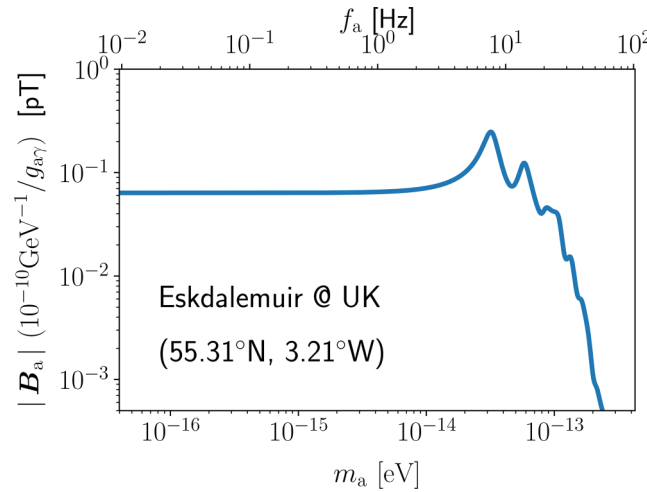


Fig. 1. Expected magnetic field amplitude induced by the coherently oscillating axion, $|B_a|$, at Eskdalemuir Observatory. The amplitude is normalized by the axion–photon coupling strength of $10^{-10} \text{ GeV}^{-1}$. The result is shown as a function of the axion mass m_a (lower) and the frequency f_a (upper) of the induced EM waves.

ionosphere. Following the treatment in the literature (e.g., Refs. [28,29]), we thus impose the boundary condition that allows only upgoing waves at the altitude of the lower atmosphere, $r \sim R_E + 100 \text{ km}$.

With the boundary conditions specified above, we solve the electric field equation in Eq. (11) by expanding it in the vector spherical harmonics, which separates the angular and radial dependences. The radial component is governed by an ordinary differential equation that admits an analytical solution when the refractive index $n(r)$ or the conductivity profile $\sigma(r)$ is divided into spatially constant segments. The solution for each segment consists of a homogeneous part and a term sourced from the axion-induced alternating current, with the coefficients of the homogeneous solution left undetermined. These are later determined by enforcing the boundary conditions and ensuring continuity between segments. Consequently, a global solution valid at all segments is constructed, and, as a result, the amplitude of the electric field becomes linearly proportional to the axion–photon coupling, $g_{a\gamma}$. More details on the construction of a solution as well as on the properties of EM fields can be found in Ref. [21].

Once the solution of the axion-induced electric field is obtained, the magnetic field is computed from Eq. (4) (assuming a monochromatic EM wave):

$$\mathbf{B} = -\frac{i}{m_a} \nabla \times \mathbf{E}. \quad (13)$$

Then, the resultant magnetic field expression is also linearly proportional to the coupling $g_{a\gamma}$, with its non-zero component appearing in the non-radial direction, that is, in the direction tangent to the Earth’s surface. In the next section, we use the measurement data of the terrestrial magnetic fields to search for the axion-induced signal. This is because the magnetic fields are easier to measure and because long-lasting data from observatories around the world are publicly available.

Figure 1 illustrates the predicted magnetic field amplitude in the non-radial mode as a function of m_a or the frequency f_a of the induced EM signal, at Eskdalemuir Observatory, UK (55.31° N , 3.21° W), where long-term monitoring data at high frequencies up to 100 Hz are available (see Sec. 3). Here, in solving the EM field equation, we used the model of the at-

mospheric conductivity profile tabulated in Ref. [26]. In Fig. 1, we assume that the axions constitute DM having the local density of $\rho_{\text{DM}} = 0.3 \text{ GeV cm}^{-3}$, with the coupling parameter of $g_{a\gamma} = 10^{-10} \text{ GeV}^{-1}$. There are prominent peaks, known as the Schumann resonances [30,31] (see also Refs. [32–35]), which are caused by EM waves trapped between the Earth's surface and the ionosphere form a resonant cavity. The first peak appears at the frequency $f_a \sim 7.8 \text{ Hz}$, corresponding to $m_a \sim 3 \times 10^{-14} \text{ eV}$.

The typical amplitude is estimated from the modified Ampère–Maxwell law, Eq. (5). Equating the terms in Eq. (5) and assuming the characteristic scale of the produced EM waves of R yield $|\mathbf{B}_a| \sim g_{a\gamma} m_a a_0 R |\mathbf{B}_{\text{geo}}|$; that is,

$$|\mathbf{B}_a| \sim 0.3 \text{ pT} \left(\frac{g_{a\gamma}}{10^{-10} \text{ GeV}^{-1}} \right) \left(\frac{\rho_{\text{DM}}}{0.3 \text{ GeV cm}^{-3}} \right)^{1/2} \times \left(\frac{R}{R_E} \right) \left(\frac{|\mathbf{B}_{\text{geo}}|}{50 \mu\text{T}} \right), \quad (14)$$

where the Earth's radius is $R_E = 6371 \text{ km}$. This estimation is consistent with Ref. [14]. On the other hand, at $m_a \gtrsim 10^{-13} \text{ eV}$, the magnetic field amplitude is sharply suppressed, reflecting the dipole nature of the geomagnetic fields, where the higher multipoles are sufficiently small. Although the amplitude in Eq. (14) is more than seven orders of magnitude smaller than that of the geomagnetic fields, it is still accessible with a high-precision magnetometer, even for commercial use. Further, the major background noise is a random superposition of transient EM waves, which can be discriminated from a persistent axion signal having a sharp spectral feature by using long-term monitoring data of terrestrial magnetic fields.

3. Data analysis method

Magnetic fields measured by a magnetometer are composed of background fields $\mathbf{B}_b$, such as the Schumann resonance and magnetometer noise, and an axion-induced signal $\mathbf{B}_a$:

$$\mathbf{B}(t) = \mathbf{B}_b(t) + \mathbf{B}_a(t). \quad (15)$$

The Fourier transform of the magnetic field is defined by

$$\mathbf{B}(t) \equiv \int df \tilde{\mathbf{B}}(f) e^{-2\pi i f t}. \quad (16)$$

Because the axion signal is proportional to the coupling constant $g_{a\gamma}$ and is assumed to be coherent and monochromatic during the observation time, we can define the normalized amplitude of the axion-induced magnetic fields, $\mathbf{R}$, and write the magnetic fields as

$$\mathbf{B}_a(t) \equiv \mathbf{R}(f_a) g_{a\gamma} e^{-2\pi i f_a t}, \quad (17)$$

where f_a is the frequency corresponding to the axion mass, m_a . We assume that $\mathbf{R}(f_a)$ is time-independent and that the axion-induced signal is stationary. Then, the Fourier component is given by

$$\begin{aligned} \tilde{\mathbf{B}}(f) &\equiv \int dt \mathbf{B}(t) e^{2\pi i f t} \\ &= \tilde{\mathbf{B}}_b(f) + \mathbf{R}(f_a) g_{a\gamma} \delta(f - f_a). \end{aligned} \quad (18)$$

Assuming that the magnetic fields are stationary in a data segment, which typically lasts for several hours, and defining the power spectra

$$\langle \tilde{\mathbf{B}}^*(f) \tilde{\mathbf{B}}(f') \rangle \equiv P(f) \delta(f - f'), \quad (19)$$

$$\langle \tilde{\mathbf{B}}_b^*(f) \tilde{\mathbf{B}}_b(f') \rangle \equiv P_b(f) \delta(f - f'), \quad (20)$$

where $\langle \cdot \rangle$ is the ensemble average, we have

$$P(f)\delta(f - f') = P_b(f)\delta(f - f') + |\mathbf{R}(f_a)|^2 g_{a\gamma}^2 \delta(f - f_a)\delta(f' - f_a). \quad (21)$$

Here we ignored a crosstalk between $\tilde{\mathbf{B}}_b(f)$ and $\mathbf{R}(f_a)$ at $f = f_a$ because there is no correlation between sources for $\tilde{\mathbf{B}}_b(f)$ and $\mathbf{R}(f_a)$. We define the single-sided power spectrum of magnetic fields as

$$\begin{aligned} S(f) &\equiv 2P(f) \\ &\approx S_b(f) + \frac{2}{T_{\text{obs}}} |\mathbf{R}(f_a)|^2 g_{a\gamma}^2 \{\delta(f - f_a)\}^2, \end{aligned} \quad (22)$$

where the factor of two in the first line arises from the contributions from positive and negative frequencies. In the second line, because the observation time T_{obs} is finite, we used $\delta(0) \approx T_{\text{obs}}$ (see e.g. Ref. [36] for the definition). Also, we replaced the double-sided power spectrum $P_b(f)$ with the single-sided power spectrum $S_b(f)$.

We divide the time series data of the magnetic fields into segments of a smaller period T_{seg} , typically several hours for stationarity of the data, and try to subtract the Schumann magnetic fields from the data at the frequency of the axion signal, $f = f_a$.

We consider an axion signal at $f = f_a$ in the i th segment:

$$S^{(i)}(f_a) = S_b^{(i)}(f_a) + 2T_{\text{seg}} |\mathbf{R}(f_a)|^2 g_{a\gamma}^2. \quad (23)$$

The power spectrum of the Schumann magnetic fields at $f = f_a$ can be estimated from the power spectrum around $f = f_a$ by smoothly interpolating it. Denoting the smooth spectrum of each segment by $\tilde{S}^{(i)}(f_a)$ and subtracting it from $S_b^{(i)}(f_a)$, which we call the i th-segment differential data, we define an estimator of the axion line signal at $f = f_a$ by summing up all differential data segments multiplied by the weights, $w^{(i)}$, based on the square inverse of the mean $S_b(f_a)$,

$$\hat{s}(f_a) \equiv s_b(f_a) + 2T_{\text{seg}} |\mathbf{R}(f_a)|^2 g_{a\gamma}^2, \quad (24)$$

where

$$s_b(f_a) \equiv \sum_i \frac{w^{(i)}}{W} \left\{ S_b^{(i)}(f_a) - \tilde{S}^{(i)}(f_a) \right\}, \quad (25)$$

$$W \equiv \sum_i w^{(i)}. \quad (26)$$

The definition of the weight $w^{(i)}$ is introduced below. Because the weighted average and the subtraction of the smooth spectrum are commutable, the spectral subtraction can be performed once after the weighted average is taken. The above analysis procedure is summarized in Fig. 2.

From the definition of the estimator, if the axion signal is sharp enough, the differential spectrum, $s_b(f_a)$, is statistically random about zero. The axion signal is always positive only if the signal amplitude is larger than the smoothed spectrum.

Writing the variance of noise as $[\Delta \hat{s}(f_a)]^2 \equiv \text{Var}[s_b(f_a)]$, from Eq. (24), we define SNR as

$$\text{SNR} \equiv \sqrt{\frac{\langle \hat{s}(f_a) \rangle}{\Delta \hat{s}(f_a)}} \quad (27)$$

$$= \sqrt{\frac{2T_{\text{seg}}}{\Delta \hat{s}(f_a)}} |\mathbf{R}(f_a)| g_{a\gamma}. \quad (28)$$

When segment data are equally weighted, that is, $w^{(i)} = 1$, the noise variance is reduced to

$$[\Delta \hat{s}(f_a)]^2 = \frac{1}{N_{\text{seg}}^2} \left\langle \sum_i \left\{ S_b^{(i)} - \tilde{S}^{(i)} \right\}^2 \right\rangle. \quad (29)$$

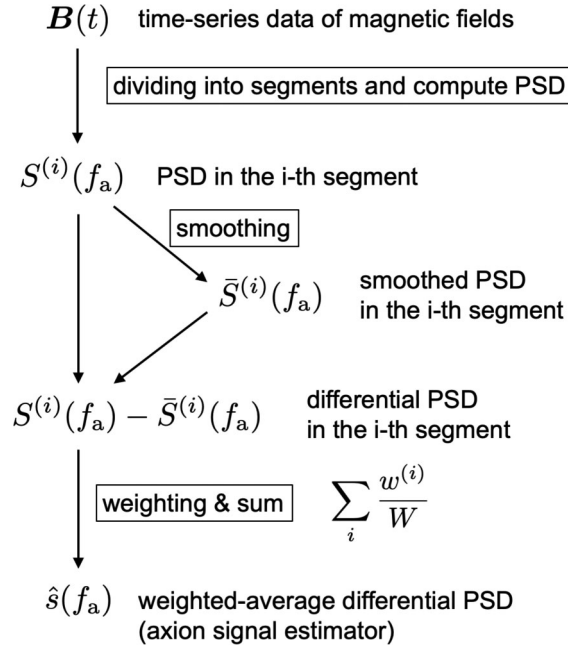


Fig. 2. Flow chart of the analysis computing the axion signal estimator.

Combining with Eq. (28), we find that the SNR is improved in proportion to $N_{\text{seg}}^{1/4}$ as the observational time increases, recovering the standard scaling [37].

4. Data and analysis results

4.1. Data

We use the publicly available magnetic field data deposited at the British Geological Survey [38]. The data were taken by two induction coils (CM11E1) at Eskdalemuir Observatory, UK [39] from September 1, 2012 to November 4, 2022. There are two channels, North–South (CH1) and East–West (CH2), with a sampling frequency of 100 Hz. The raw data are given in digitizer units and need to be converted to picotesla (pT) using the conversion factors 3.491×10^{-6} V/count for CH1 and 3.475×10^{-6} V/count for CH2 and the frequency-dependent calibration factors between 10^{-3} and 10^2 Hz in Table A1 in Appendix A. The response of the magnetometer decreases at low frequencies, by 5% at 10^{-2} Hz and by more than 90% at 10^{-3} Hz. For the reason, we set the lowest frequency for the use of the data to 10^{-3} Hz and the highest frequency to the Nyquist frequency, 50 Hz. We also checked the absolute calibration by comparing the data with those measured with a different instrument at Eskdalemuir Observatory and found that the data we use are well calibrated throughout the period between 2012 and 2022. The details of the comparison are provided in Appendix A.

4.2. Estimator for axion signals

For the purpose of searching for axion signals, the orientation of the measured magnetic fields is helpful to distinguish the signal from noise [21]. The orientation angle error of the Eskdalemuir coil data is of the order of a few degrees [38], but the calibration information is not provided. For a conservative analysis, we decided to use only the signal power spectral density (PSD), computed from the two channels, CH1 and CH2. Furthermore, to search for a sharp line signal, stably long-lasting data are necessary. Therefore, we use only data lasting more than one

Table 1. The amount and quality of data used for the analysis. The number of data segments is counted in the unit of 8 h. $w^{(Y)}$ is the sum of $w^{(i)}$ in each year, and W is the sum of $w^{(Y)}$.

Year	Number of data segments	$w^{(Y)}/W$
2012	176	0.0247
2013	1028	0.1137
2014	581	0.0455
2015	1007	0.0684
2016	1097	0.0962
2017	976	0.0766
2018	984	0.1483
2019	968	0.1126
2020	1095	0.1022
2021	1094	0.1532
2022	903	0.0595
all years	9909	1

month. Then, each sequential data are divided into 8-h segments. The length of a data segment is chosen so that the frequency width of an axion signal at 35 Hz, that is, $\Delta f_a \sim (v_{\text{DM}}/c)^2 f_a \sim 3.5 \times 10^{-5}$ Hz, is included in a single frequency size of $1/(8 \text{ h}) \approx 3.5 \times 10^{-5}$ Hz. As the geomagnetic fields are rapidly damped above 30 Hz, the choice of frequency bin size hardly impacts the sensitivity above 30 Hz. Axion signals at frequencies below 30 Hz are well within a single frequency bin. This is not an optimal choice, considering the coherent time of axion DM, particularly at lower frequencies. However, there are 30 second-long transient noises with huge amplitude (saturated at $\sim 10^8$ pT). These prevent us from taking longer data segments. To avoid contamination from the transient noise, we fix the data segment size to 8 h. The number of 8-h data segments that we used for our analysis is shown in Table 1. The total number of the data segments is $N_{\text{seg}} = 9909$.

Given the PSD of the magnetic fields for each data segment, we compute the estimator of an axion signal, $\hat{s}(f_a)$ in Eq. (24). We stack the PSDs multiplied by the weight $w^{(i)}$, which is defined by the inverse square of the mean $S_b(f_a)$ computed from the spectrum between 1.1 Hz and 9.9 Hz², and then subtract the smooth component obtained by applying a quadratic Butterworth filter with the cutoff frequency of 2.5 Hz. Dividing by the sum of the weights for data segments, W , we obtain the weighted average differential PSD, $\hat{s}(f_a)$. With these procedures, the sharp line of an axion signal is constructively stacked and the spectral features of the Schumann resonance are subtracted (but not perfectly).

As written in the document on the data server [38], at the integer-valued frequencies, there are artificial noise lines that appear irregularly and whose cause is unknown. We veto those artificial line noises by removing the data in the range of $\pm 3.47 \times 10^{-2}$ Hz (± 1000 frequency bins) below and above the integer-valued frequencies.

The weighted-average differential PSD, $\hat{s}(f_a)$, is plotted in Fig. 3. The spectral amplitude (blue) increases toward low frequencies below 1 Hz at the timescales of storms, and is almost flat between 1 and ~ 100 Hz with Schumann resonance peaks [40]. The second and third panels from the top are the same as the first one but in different frequency ranges. Spectral lines are

²The frequency range is chosen to average a flat spectrum in the most sensitive frequency band up to 10 Hz (see Fig. 1).

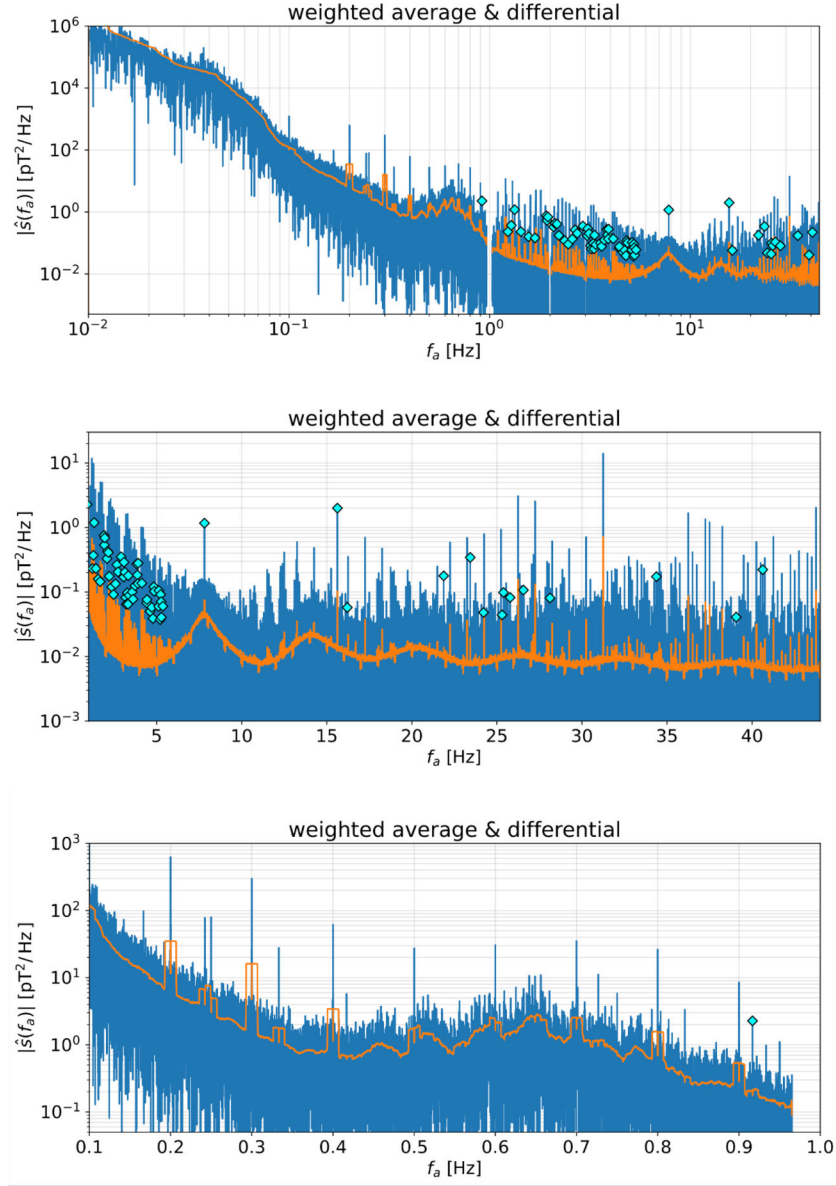


Fig. 3. Weighted-average differential PSD of the magnetic fields, stacked over all 8-h data segments with noise-based weights and then subtracted the smooth spectrum, $\hat{s}(f_a)$ in blue. The standard deviation is depicted in orange. The axion signal candidates are marked by diamonds in cyan. The second and third panels from the top are the same as the first one but in different frequency ranges.

concentrated above 1 Hz. Below 1 Hz, prominent lines appear at the frequencies of multiples of 0.1 Hz.

4.3. SNR

The SNR is computed from Eq. (27). Note that the definition of the SNR is sensitive to the sign of the weighted average differential PSD, which plays a crucial role when we check the persistency of the signals. The standard deviation of noise, $\Delta\hat{s}(f_a)$, is evaluated from the data of ± 200 frequency bins ($\pm 3.47 \times 10^{-3}$ Hz) around $f = f_a$, except for three frequency bins centered at $f = f_a$, which is shown in orange in Fig. 3. With this treatment, the noise variance estimation does not include an axion signal candidate itself and automatically excludes line signals broader

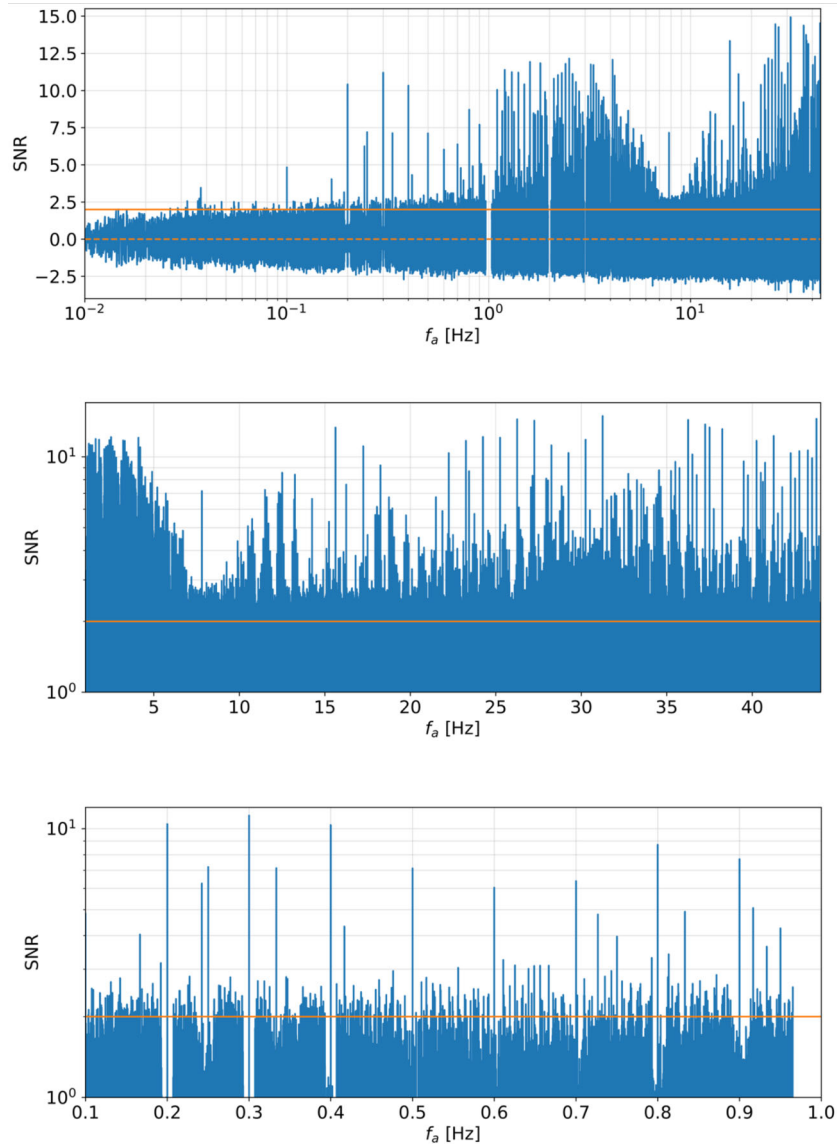


Fig. 4. SNR with 8-h data segments from all-year data. In the three panels, the same data are plotted in the different ranges of frequencies corresponding to axion masses. The horizontal solid and dashed lines are $\text{SNR} = 2$ and $\text{SNR} = 0$. The second and third panels from the top are the same as the first one but in different frequency ranges.

than a single frequency bin. For the procedure, we set the lower limit of f_a to 0.01 Hz to have the sufficient number of frequency bins for noise variance evaluation, while the upper limit of f_a is set to 44 Hz because the spectrum around and above 50 Hz is unreliable owing to the Nyquist frequency.

In Fig. 4, the SNR as a function of the frequency corresponding to axion mass is plotted. The second and third panels from the top are the same as the first one but in different frequency ranges. The SNRs are likely to be larger at higher frequencies because the noise variance is smaller there. The signals above the SNR threshold are concentrated above 1 Hz, particularly in the range of 1–6 Hz.

Depending on the SNR threshold, we found a number of signal candidates, listed in the second row in Table 2. Filtering $\text{SNR} > 3$ for all-year data leaves 821 signal candidates. The can-

Table 2. Number of signal candidates filtered by the conditions of the SNR threshold, persistency, and not-neat frequency.

SNR threshold	All years	And persistency	And excluding neat frequencies
SNR > 14	5	5	0
SNR > 13	9	9	1
SNR > 12	14	14	1
SNR > 11	32	31	1
SNR > 10	46	41	1
SNR > 9	60	49	2
SNR > 8	96	74	9
SNR > 7	132	99	13
SNR > 6	183	131	22
SNR > 5	257	172	31
SNR > 4	473	270	50
SNR > 3	821	375	65
SNR > 2	26912	660	342

didates are mostly in a single frequency bin and are consistent with the sharpness of an axion signal. If a signal candidate is true, it must be persistent through all the data. We further set the condition for the candidates to be credible, requiring that the SNR for each-year data is above the weighted SNR threshold, $\text{SNR} \times w^{(Y)}/W$, where $w^{(Y)}$ is the sum of $w^{(i)}$ in each year and is listed in Table 1. The numbers of signal candidates after imposing the condition of persistency are also listed in the third row in Table 2. Persistent filtering works efficiently for candidates with low SNR and reduces the number of candidates drastically, because noise may change its sign every year owing to the subtraction of the smooth component in Eq. (25) but an axion signal does not change. The second condition reduces the number of signal candidates from 821 to 375 for $\text{SNR} > 3$.

To reduce the number of candidates further, we rule out candidates at frequencies of multiples of 0.05 Hz (within 0.001 Hz), called “neat frequencies”. These signals are likely to be produced by the analogue filters or the digitizer [41]. After excluding the candidates at the neat frequencies, the number of candidates is significantly reduced to 65 for $\text{SNR} > 3$.

In the selection of signal candidates above, we did not take into account the look-elsewhere effect, which is the effect of finding more signal candidates when more trails are performed or, in our case, more frequency bins exist. Considering the look-elsewhere effect, the p -value threshold for the identification of a candidate, e.g. [14,15], is $p_{\text{th}} = 1 - (1 - p_1)^{N_f}$, where p_1 is the p -value threshold when a single frequency bin is analyzed and N_f is the number of frequency bins analyzed. In our case, $N_f = 1.3 \times 10^6$. Given $p_1 = 0.05$, we have $p_{\text{th}} = 4.0 \times 10^{-8}$, corresponding to $\text{SNR} = 5.5$ for a Gaussian noise distribution. Then we have 25 signal candidates with $\text{SNR} > 5.5$.

Considering the look-elsewhere effect reduces the number of signal candidates that happen to be caused by noise fluctuations. However, it may discard a true signal with modest SNR in the current search. To be conservative, we keep all the signal candidates found with $\text{SNR} > 3$ in this search. The signal candidates with $\text{SNR} > 3$ are listed in Table 3 and are shown with the filled diamonds in Fig. 3. They are concentrated above 1 Hz, particularly in the range of 1–6 Hz.

Table 3. Frequencies and SNRs of persistent significant signal candidates above SNR= 3 for all-year data and above the SNR threshold weighted by noise level for each-year data, removing the candidates of neat frequencies.

ID	Frequency [Hz]	SNR	ID	Frequency [Hz]	SNR	ID	Frequency [Hz]	SNR
1	0.9167	5.071	23	3.1333	6.766	45	5.1167	3.941
2	1.2333	3.584	24	3.1833	4.347	46	5.1333	5.109
3	1.2833	4.696	25	3.2167	3.996	47	5.2167	3.002
4	1.3333	8.580	26	3.2833	4.813	48	5.2333	4.642
5	1.4333	4.184	27	3.3167	4.164	49	5.2667	4.110
6	1.5667	3.767	28	3.3333	6.571	50	5.2833	3.076
7	1.6833	3.736	29	3.3833	4.637	51	5.3667	3.825
8	1.9167	9.374	30	3.5667	5.240	52	7.8125	7.168
9	1.9333	8.288	31	3.6167	4.581	53	15.6250	13.342
10	1.9531	6.336	32	3.7333	5.837	54	16.2052	3.105
11	2.0833	6.867	33	3.8333	8.211	55	21.8750	5.896
12	2.1167	7.939	34	3.8667	6.892	56	23.4375	8.704
13	2.1667	7.787	35	3.9062	4.317	57	24.2188	3.459
14	2.2167	5.234	36	3.9167	8.036	58	25.3073	3.052
15	2.3667	4.536	37	4.1333	6.148	59	25.3906	4.468
16	2.4667	4.154	38	4.4167	4.306	60	25.7812	4.018
17	2.5833	5.059	39	4.4333	4.589	61	26.5625	4.487
18	2.6667	3.375	40	4.6167	3.617	62	28.1250	4.336
19	2.7333	6.445	41	4.7167	4.008	63	34.3750	6.699
20	2.9167	8.538	42	4.7833	3.245	64	39.0625	3.169
21	3.0667	6.273	43	4.8333	5.596	65	40.6250	7.852
22	3.0833	8.200	44	4.8667	5.200			

The highest SNR signal has SNR= 13.342 and is the only signal above the SNR threshold of 10. There are many candidates at multiples of 0.33 and 0.0033 Hz. Although these frequencies seem to be related to an artificial cause, we have no information about the possible cause [41] and keep them as candidates to be conservative.

4.4. Axion signal candidates

For further investigation of the axion signal candidates, we plot in Fig. 5 the top-four-SNR axion signal candidates and four large-SNR noise-like candidates at neat frequencies. They are in one or two frequency bins, consistent with the frequency width of the axion signal, and have no remarkable difference in spectral features. There are shallow dips around the sharp lines. This is because a smoothed spectrum is computed by applying a quadratic Butterworth filter with the cutoff frequency of 2.5 Hz and is subtracted from the original spectrum in Eq. (25). Note that noise variance does not differ significantly for the signal candidates, but the ranges of the vertical axis are different, depending on their SNRs.

As mentioned in the previous section, the candidates at neat frequencies are likely to be produced by the analogue filters or the digitizer [41]. Because we have no additional information to veto the axion signal candidates further, all the signal candidates that pass our three signal criteria should be kept as possible axion signals for future searches. On the other hand, one should investigate the signal candidates deeply by studying their noise properties or should check their consistency with the magnetic field data at other observatories.

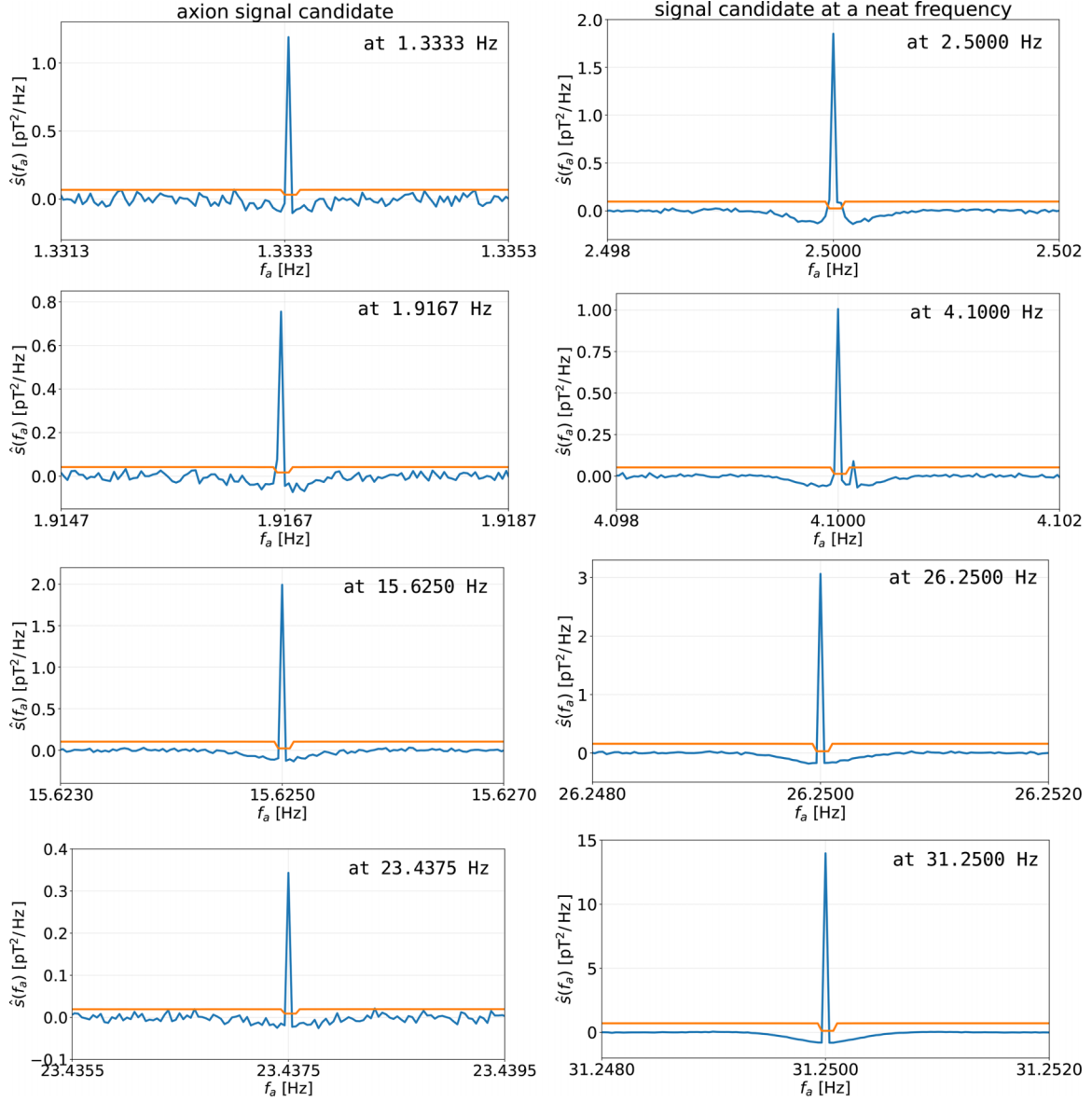


Fig. 5. Axion signal candidates at 1.3333 Hz (SNR= 8.580), 1.9167 Hz (SNR= 9.374), 15.6250 Hz (SNR= 13.342), and 23.4375 Hz (SNR= 8.704), and the signal candidates at neat frequencies, 2.5000 Hz (SNR= 12.152), 4.1000 Hz (SNR= 12.073), 26.2500 Hz (SNR= 14.460), and 31.2500 Hz (SNR= 14.932). The left-hand panels are axion signal candidates with the top-four SNRs, and the right-hand panels are the significant signal candidates rejected owing to neat frequencies. The signal, $\hat{s}(f_a)$, is in blue, and its standard deviation as noise is in orange.

In Table 4, we list the signal candidates with $\text{SNR} > 2$ at low frequencies (below 1 Hz). In the previous study using SuperMAG 1-s sampling data [15], there were three signal candidates with modest significance. However, none of the candidates in Table 4 is identified with those found in the previous search. Although our sensitivity is slightly worse than that in the previous study at 0.2630 Hz, which is one of the signal candidate frequencies in the previous search, we did not find any signal candidate at the frequency and constrain the existence of axions with coupling strength larger than $g_{a\gamma} \lesssim 1.3 \times 10^{-11} \text{ GeV}^{-1}$ at the 95% credible level (CL).

Table 4. Frequencies and SNRs of persistent signal candidates above SNR = 2 (below 1 Hz) for all-year data and above the SNR threshold weighted by noise level for each-year data, removing the candidates of neat frequencies.

ID	Frequency [Hz]	SNR
A	0.4339	2.030
B	0.4764	2.957
C	0.5275	2.364
D	0.8851	2.626
E	0.9167	5.071

5. Observational constraint

Except for 342 frequency bins at which signal excesses with SNR > 2 were found, the data are consistent with noise, and we obtain the upper limit on the axion coupling strength in the frequency range of 0.01 – 43.97 Hz. The 95% CL upper limit on the axion–photon coupling strength is obtained from

$$\int_{-\infty}^{\hat{s}_{\text{obs}}(m_a)} d\hat{s} p[\hat{s}(m_a)|g_{a\gamma}] = 0.05, \quad (30)$$

where $\hat{s}_{\text{obs}}$ is the observed value of the weighted-average differential PSD, which can be converted to the axion coupling strength by Eq. (24) and the theoretical curve from Fig. 1. $p[\hat{s}(m_a)|g_{a\gamma}]$ is the probability density distribution of the weighted-average differential PSD in the presence of axions, which is obtained from the data $\hat{s}$ of the surrounding ± 200 bins, excluding the three bins around $f = f_a$ and adding an axion signal in Eq. (24) at $f = f_a$. Because the frequency bin of the axion signal and the frequency bins of noise sampling do not overlap, they do not affect each other. As shown in Appendix B, the distribution of the noise amplitude, $\hat{s}$, is well approximated by the Gaussian distribution with zero mean³, except for the frequency range of 1.1–3.9 Hz, indicating that the 95% CL upper limit obtained in this analysis roughly corresponds to twice the standard deviation of the Gaussian distributions.

The upper limit on the axion–photon coupling is shown in Fig. 6. Around the axion mass of 3×10^{-14} eV, the upper limit is the tightest and is improved from that by CAST [16,17] by about two orders of magnitude and exceeds that obtained from astrophysical X-ray observations by Chandra [42,43] and NuSTAR [44]. In the lower mass range, our constraint is weakened because of fixing the data segment size to 8 h and using a broader frequency resolution, irrespective of the coherent time of axions. The constraint can be stronger by optimizing the data segment size or the frequency bin resolution for each axion mass and could be comparable to the upper limit from SuperMAG [14,15]. In the higher mass range above 10^{-14} eV, the constraint is also weakened owing to the suppression of the magnetic field response to axions. Interestingly, our new constraint closes the window between the excluded regions from SuperMAG and the Planck and unWISE blue galaxy sample [45], though the astrophysical model-dependent constraints from Chandra [42,43] and NuSTAR [44] already exist. We emphasize that the astrophysical upper limit from Chandra and NuSTAR strongly depends on the modeling assumptions and strength of intergalactic magnetic fields in the line of sight. It is difficult to estimate their uncertainty, but the strongest signal candidate (SNR ~ 13) would be compatible with the previous

³Rigorously, the distribution is a shifted chi-squared distribution. But it can be well approximated by a Gaussian distribution owing to a sufficient number of samples.

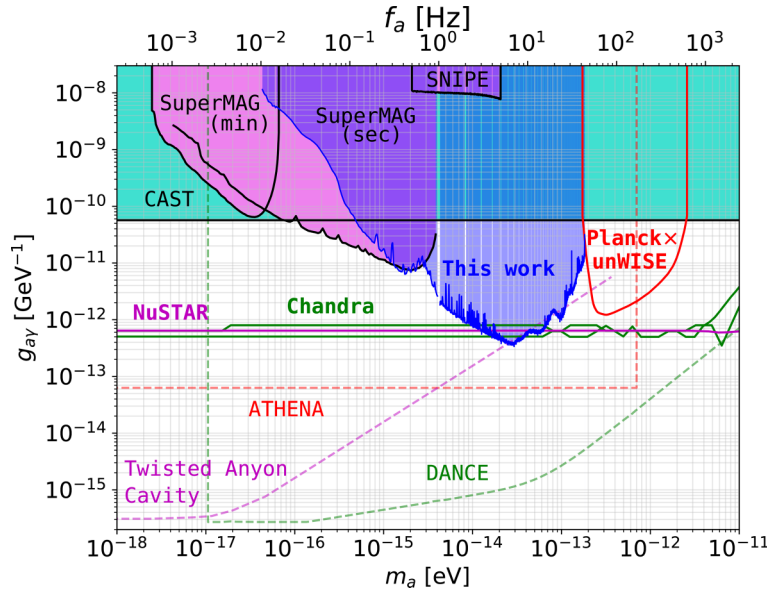


Fig. 6. Constraint (95% CL) on the axion–photon coupling $g_{a\gamma}$ from the long-term monitoring data of magnetic fields at Eskdalemuir Observatory (this work, blue). Note that our analysis does not constrain signals within 0.001 Hz of the neat frequencies (multiples of 0.05 Hz), though they are invisible in the plot. Other excluded regions are from SuperMAG [14,15], SNIPE [18], the Planck and unWISE blue galaxy sample [45], CAST [16,17], Chandra [42,43], and NuSTAR [44]. The future constraints are from Athena (red) [50], the DANCE ring cavity (green) [48], and the twisted anyon cavity (purple) [49]. The data are taken from [58]. The constraints by direct means are filled, and those by astrophysical model-dependent means are not filled.

observations [42–44] if the upper limit were weaker by an order of magnitude. For other typical candidates with an SNR of several, they are consistent with the previous upper limits [42–44] within a factor of several in the coupling strength.

Finally, we comment on the so-called stochastic effect [46,47], which is caused by random realizations of the axion field amplitude during the measurement time and could loosen the upper limit on the axion coupling. This effect is relevant when the measurement time is shorter than the coherence time of axions. In our analysis, we computed the axion field amplitude from the average density of DM and assumed the amplitude to be constant in time. However, because the measurement time is more than 10 times longer than the coherent time in the sensitive range of axion mass, the stochastic effect is negligible [37]. More quantitatively, we use 9909 8-h data segments, and the fraction of the good quality data is $\sim 10\%$, resulting in the effective measurement time of ~ 300 days. In the mass range heavier than 2×10^{-15} eV or frequency higher than 0.4 Hz, the condition above is satisfied and the stochastic effect is negligible. Therefore, our constraint improved from SuperMAG and CAST in the mass range above 2×10^{-15} eV is not affected by the stochastic effect.

6. Conclusions

In this paper, we utilized terrestrial EM waves in extremely low-frequency bands as a powerful probe for ultralight axion DM. We searched for the characteristic spectral feature induced by coherently oscillating axions interacting with the Earth’s magnetic fields in the long-term monitoring data of terrestrial magnetic fields. For the SNR threshold of 5.5 and 3 when the look-elsewhere effect is or is not considered, we found 25 and 65 axion signal candidates, respectively,

in the frequency range of 0.01 – 43.97 Hz. These candidates cannot be vetoed further owing to the lack of additional information to identify their causes. At other frequencies at which signal candidates were not found, using the theoretical prediction of the induced EM waves that properly takes the finite conductivity of the atmosphere into account, we placed the most stringent upper bound on the axion–photon coupling strength at $10^{-15} \text{ eV} \lesssim m_a \lesssim 10^{-13} \text{ eV}$ among the direct observational means.

Follow-up measurements by future ground-based experiments such as DANCE [48] and twisted anyon cavities [49] or X-ray astronomical observations by Athena [50], whose sensitivity curves are shown in Fig. 6, will play a crucial role to confirm or disprove the signal candidates. In this paper, we focused on axions as a representative ultralight DM, but the methodology for data analysis can be extended to other types of ultralight DM coupled with EM fields. Examples worth considering are dark photon DM [51,52], whose kinetic mixing has been constrained with terrestrial magnetic fields [53–55], and millicharged DM [56].

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Appendix A. Calibration of the data

The induction coil system at Eskdalemuir Observatory, UK [39] is very old. The instruments had been working since the 1970s and the digitizer since the 1990s. The calibration factors provided at the website of the British Geological Survey [38] are listed in Table A1. However, they were measured long ago and have not been checked during the period of the data we use and until the instruments finally broke in August 2023. For the reliability of the data, we investigated if the calibration factors were still correct during the period by comparing the magnetic field amplitude, $m(f) \equiv \sqrt{S(f)}$, where $S(f)$ is defined in Eq. (22), with that measured by a different instrument at Eskdalemuir Observatory in the same period. However, there is no independent high-frequency (100 Hz) sampling data at Eskdalemuir Observatory. We instead used the 1-min (0.017 Hz) sampling data taken from INTERMAGNET [57] and compared the spectral amplitude. First, we picked a day each year and compared the amplitude from 2013 to 2021 (the data on January 1 are missing in 2014 and 2022). The spectral amplitude was computed by dividing the 1-day data into 24 1-h data segments and taking the 3/24-quantile amplitude from the minimum. This was determined by the trade-off between two factors. The lower quantile is more affected by noise fluctuations and the spectrum is not smooth. The higher quantile is likely to be affected by transient noises and the spectrum happens to be large. The comparison of the 3/24-quantile amplitudes is shown in Fig. A1. They are consistent with each other, verifying that the calibration factors are correct. Second, for completeness, we compared the amplitudes in each month (a few months are missing owing to the non-existence of data) during the later period (2020–2022) of the data that we used for the axion search. As shown in Figs. A2–A4, they are consistent with each other, again verifying that the calibration factors are correct.

Table A1. Calibration factors converting nT to mV.

Frequency (Hz)	Calibration factor [mV/nT]
100	50.151
50	50.282
32	50.258
18	50.223
10	50.209
5.6	50.204
3.2	50.199
1.8	50.200
1.0	50.169
0.56	50.167
0.32	50.157
0.18	50.139
0.10	50.077
0.056	49.939
0.032	49.540
0.018	48.237
0.010	44.639
0.007	40.402
0.005	34.041
0.003	22.464
0.002	13.736
0.001	4.530

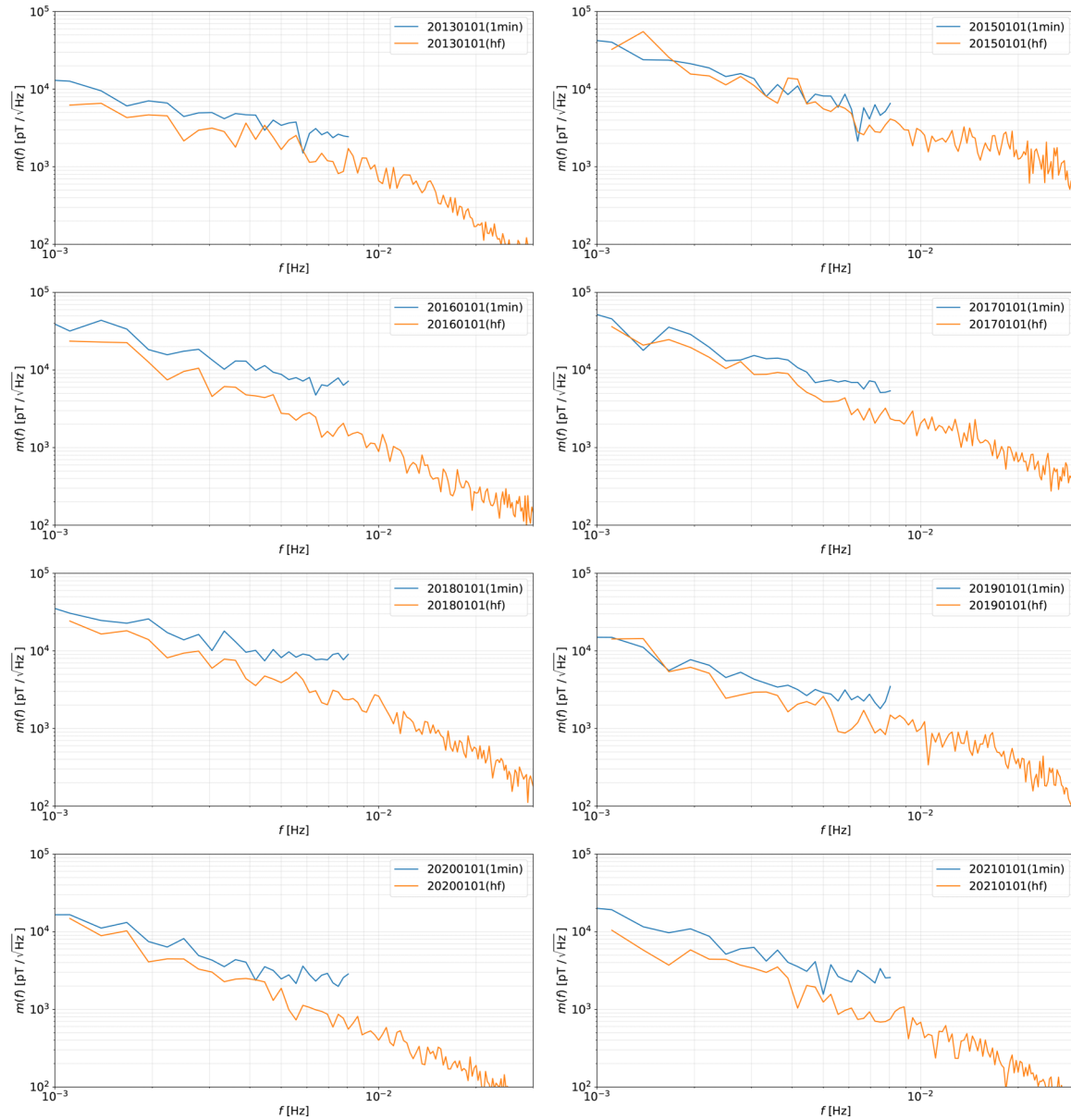


Fig. A1. Comparison of INTERMAGNET 1-min sampling data (blue) and 100-Hz sampling data from the British Geological Survey (orange) at Eskdalemuir Observatory. Each data point is the 3/24-quantile amplitude from the minimum among all 1-h data on January 1 of each year (to avoid transient loud noises).

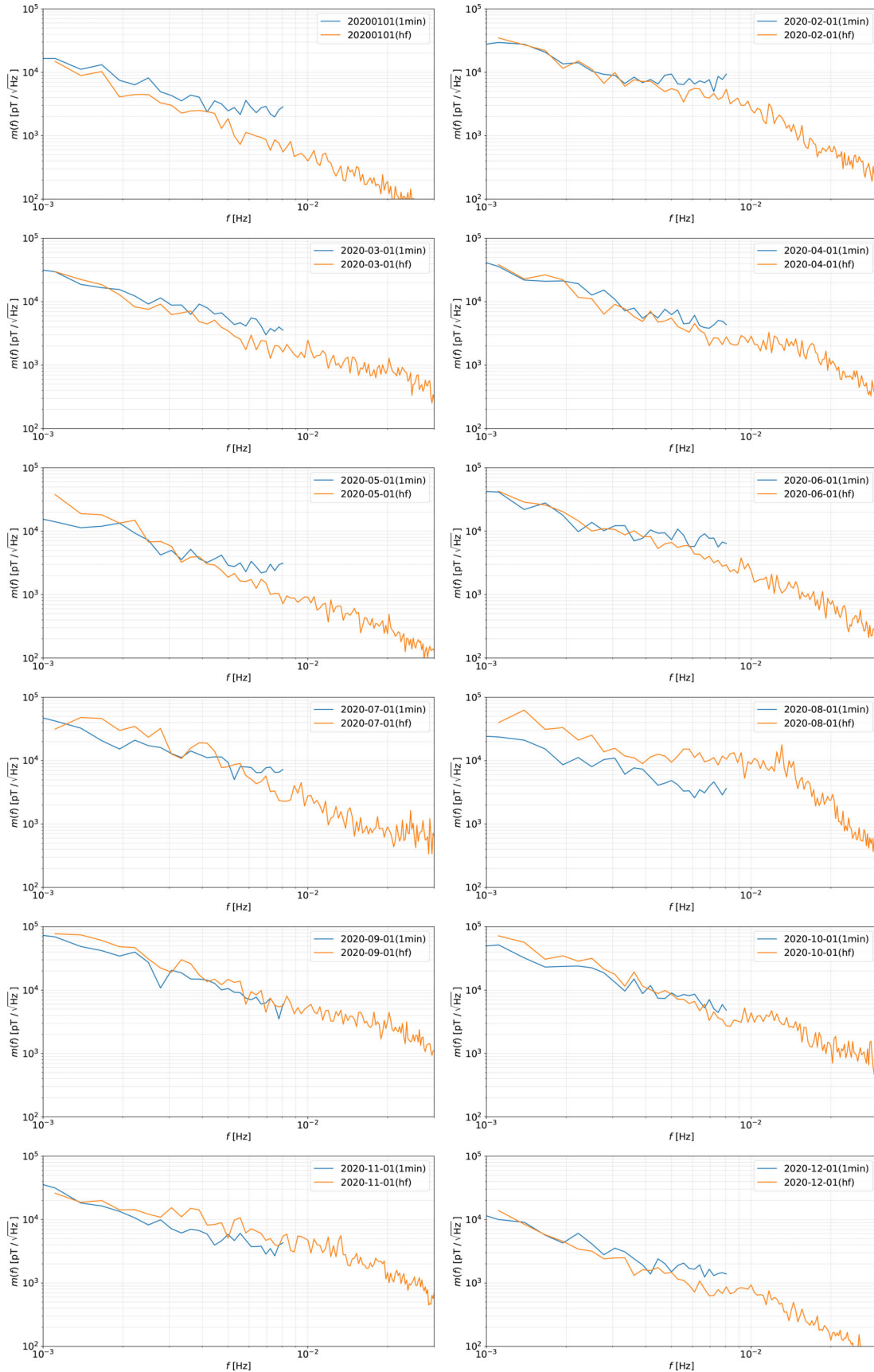


Fig. A2. Same as Fig. A1, but the data are from one day in each month, 2020.

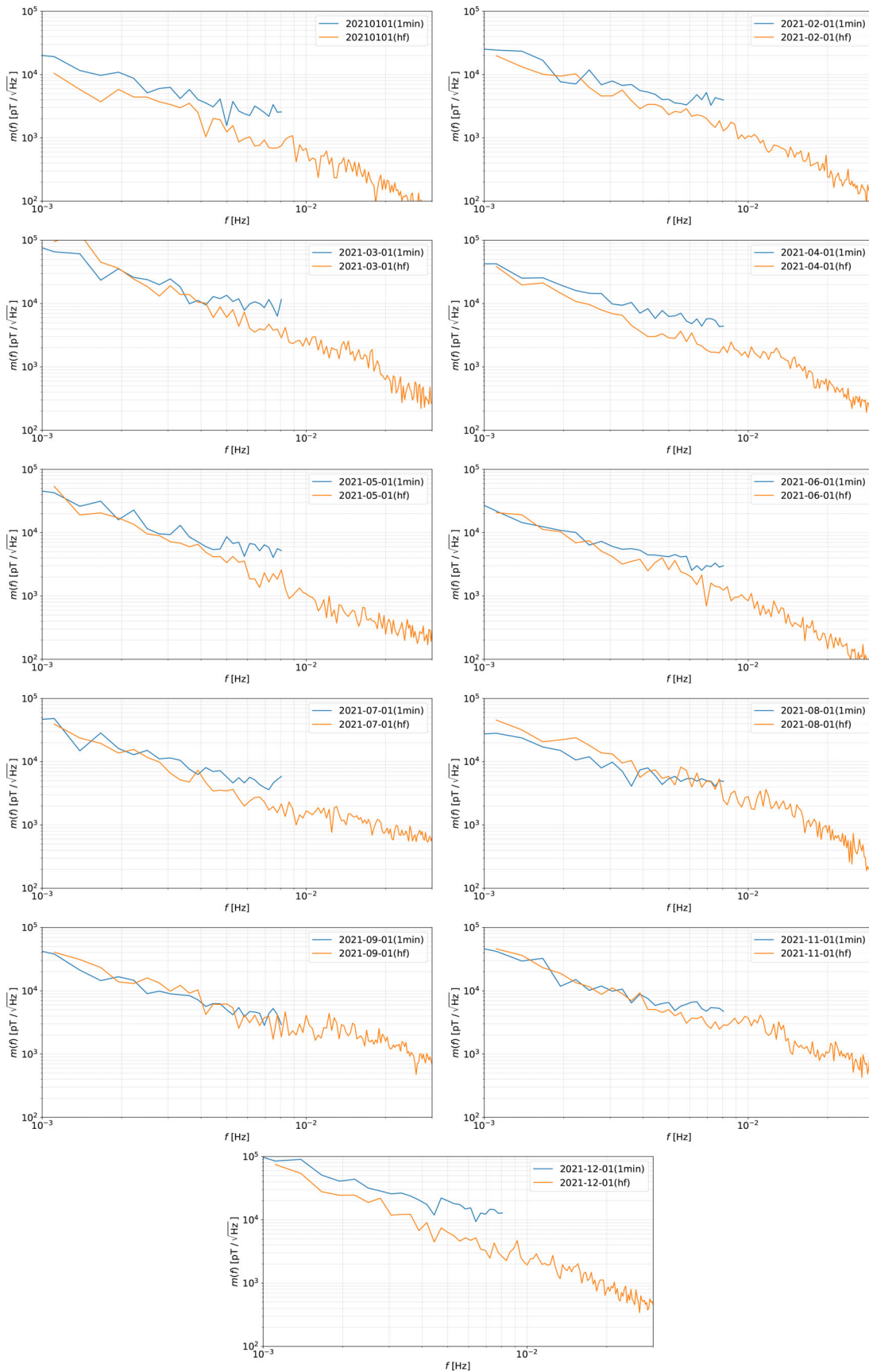


Fig. A3. Same as Fig. A1, but the data are from one day in each month, 2021.

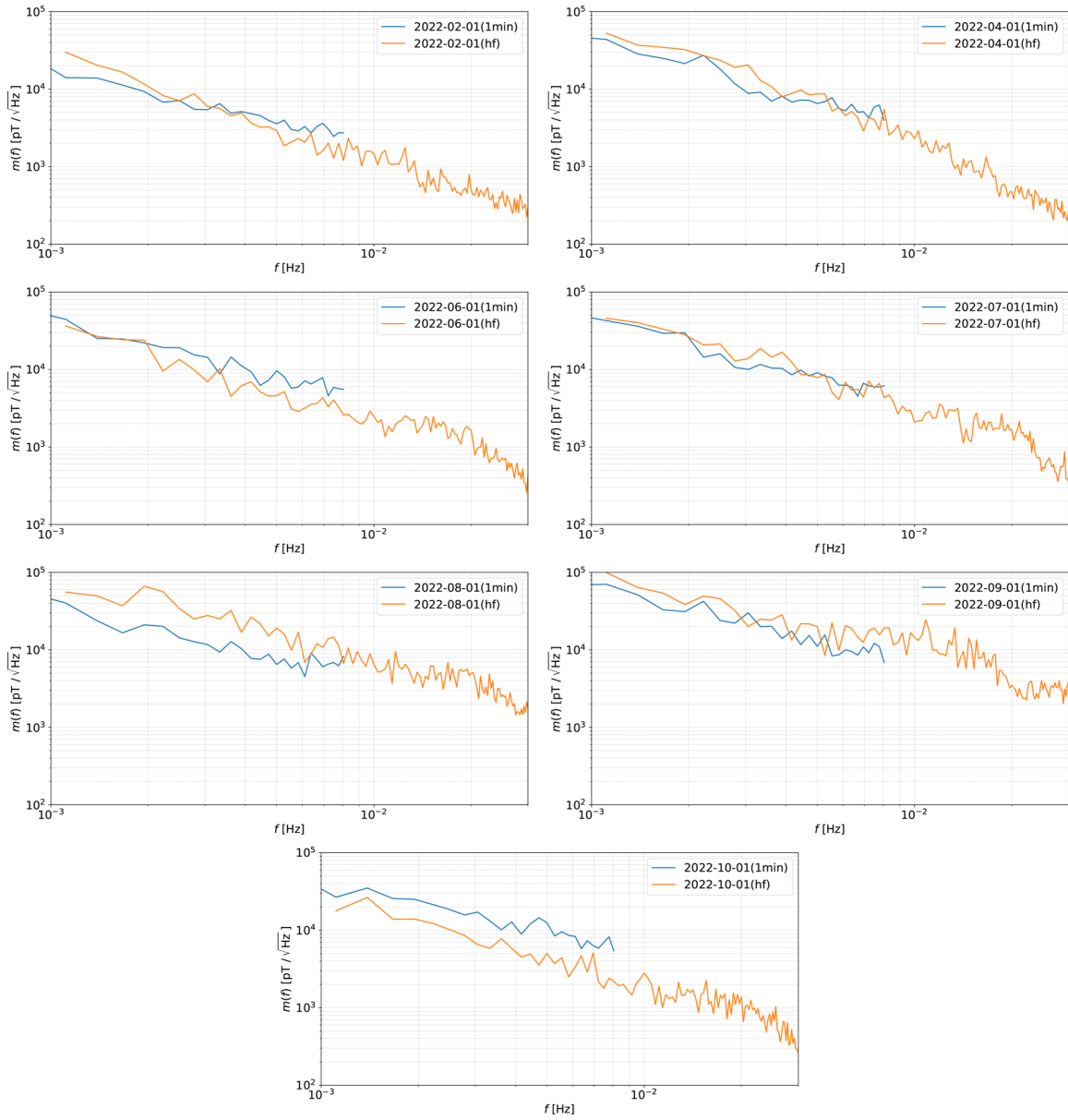


Fig. A4. Same as Fig. A1, but the data are from one day in each month, 2022.

Appendix B. Noise amplitude distribution

Noise properties are crucial to further investigate axion signal candidates and also to obtain an upper limit on the axion coupling strength. Here we show noise amplitude distributions in the low-frequency and high-frequency ranges. In Figs. B1 and B2, the distributions of the noise amplitude are plotted in several frequency intervals at low and high frequencies, respectively. The distributions are well approximated by the Gaussian distribution with zero mean except for the frequency range of 1.1–3.9 Hz. Particularly, at frequencies above 5.1 Hz, the distributions are almost indistinguishable from the Gaussian distribution. In 1.1–3.9 Hz, there are many strong lines on the spectrum, which broaden the width of the Gaussian distribution when fitted.

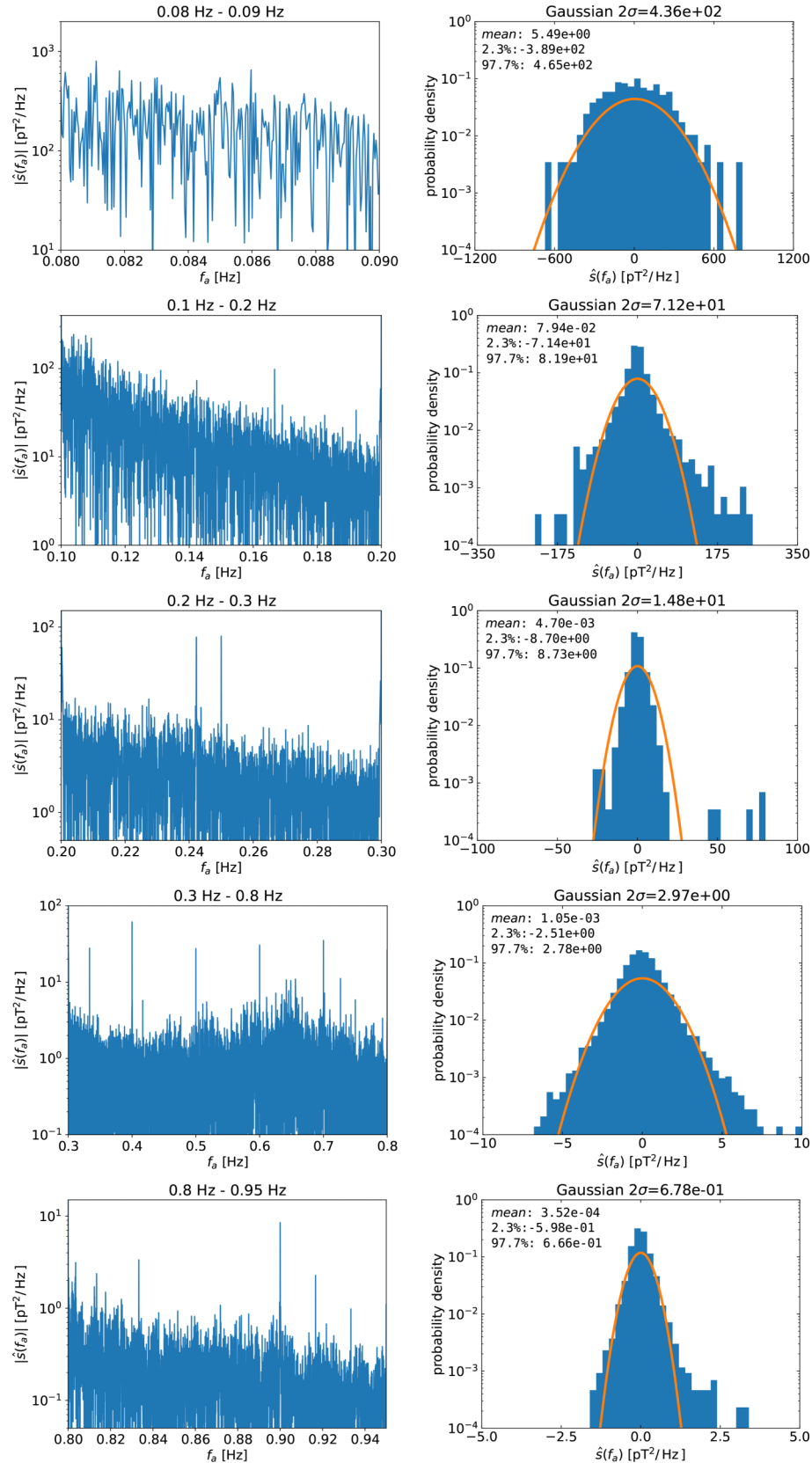


Fig. B1. Noise amplitude distributions at low frequencies. The three numbers on the right-hand panels are the mean and the 2.3% and 97.7% of the cumulative distribution of $\hat{s}(f_a)$. The numbers above the right-hand panels are twice the standard deviation when the distributions are fitted by the Gaussian distribution.

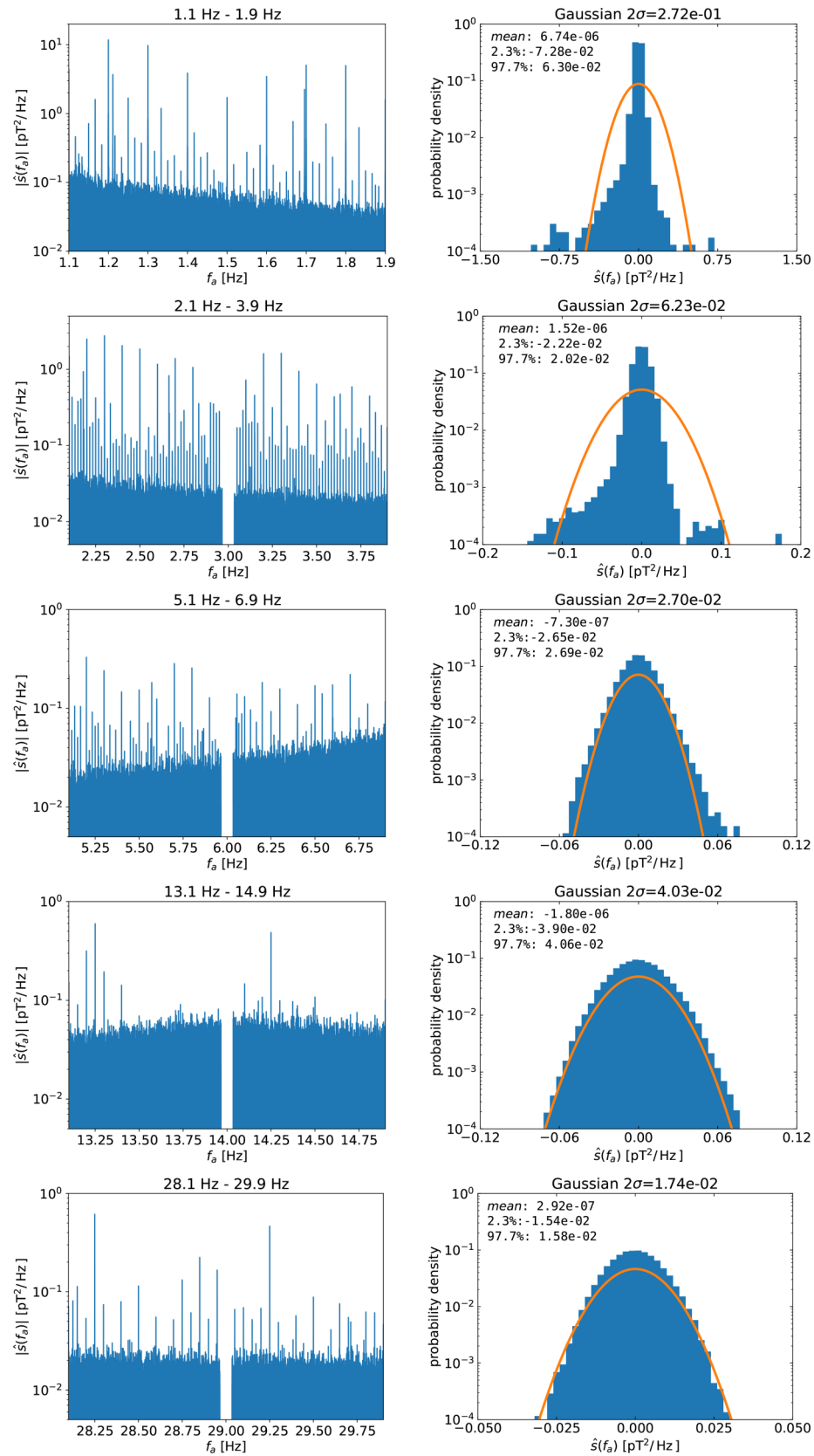


Fig. B2. Same as Fig. B1, but at high frequencies.

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